# Project Design Phase-II
# Data Flow Diagram & User Stories

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| Date | 31 January 2026 |
| Team ID | |
| Project Name | EV Market Performance and Insights |
| Maximum Marks | 4 Marks |

**Data Flow Diagrams:**

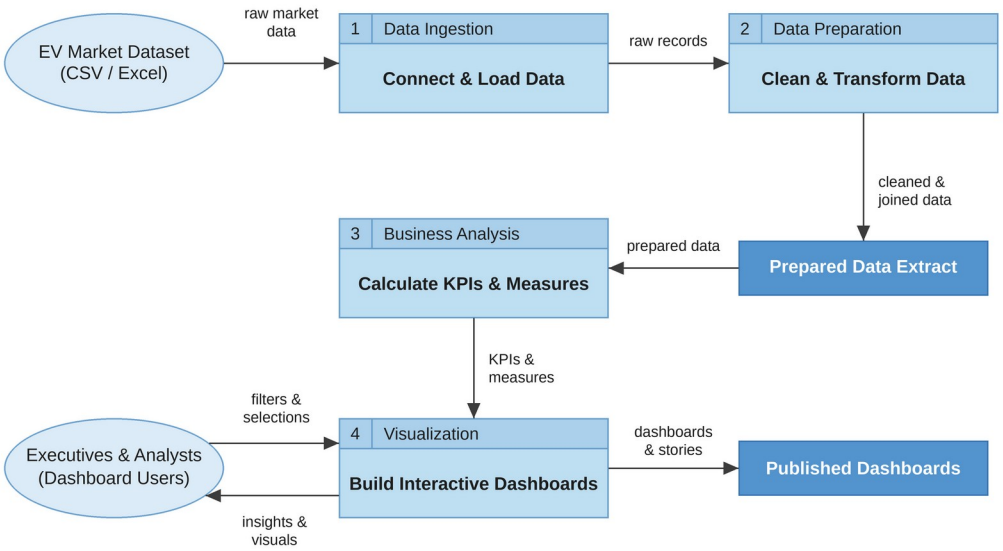
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

**DFD Symbol Legend**

| Symbol | Name | Description |
|---|---|---|
| **Oval / Rounded shape** | **External Entity** | A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway). |
| **Rectangle with numbered header** | **Process** | An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order). |
| **Rectangle (solid fill, no number)** | **Data Store** | A place where data is held for later use (e.g., Customer Database, Inventory). |
| **Labeled arrow** | **Data Flow** | The movement of data between entities, processes, and data stores, labeled with what data is |

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| | | being passed. |

DFD Level 0: EV Market Performance and Insights



## User Stories

Use the below template to list all the user stories for the product.

| User Type | Functional Requirement (Epic) | User Story Number | User Story / Task | Acceptance criteria | Priority | Release |
|---|---|---|---|---|---|---|
| Market Strategy Director | Market Performance Overview | USN-1 | As a market strategy director, I can view total EV sales, average vehicle price, average range and CO2 saved as KPI cards, so that I can assess overall market performance at a glance. | KPI cards on Dashboard 1 show correct values and respond to filters | High | Sprint-1 |
| Market Strategy Director | Market Performance Overview | USN-2 | As a market strategy director, I can see annual sales units by vehicle category from 2017 to 2027, so that I can identify high-growth segments. | Sales trend chart shows every vehicle category by model year | High | Sprint-1 |
| Market Strategy Director | Customer Satisfaction | USN-3 | As a market strategy director, I can compare top manufacturers by customer review and sales units, so that I can judge manufacturer performance and customer satisfaction. | Chart compares manufacturers by customer review and annual sales units | Medium | Sprint-2 |
| Business Intelligence Manager | Manufacturer Analysis | USN-4 | As a business intelligence manager, I can compare manufacturers by annual sales and market share, so that I can identify competitive advantages. | Manufacturer matrix ranks manufacturers by annual sales units | High | Sprint-1 |
| Business Intelligence Manager | Production Analysis | USN-5 | As a business intelligence manager, I can view production output by country and vehicle category, so that I can understand geographic production trends. | Stacked bar chart shows annual sales units by country, split by vehicle category | High | Sprint-1 |

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|---|---|---|---|---|---|---|
| Business Intelligence Manager | Battery Technology Analysis | USN-6 | As a business intelligence manager, I can see which battery technologies lead in annual sales, so that I can advise clients on product positioning. | Chart shows annual sales for LFP, NCA, NMC, Sodium-Ion and Solid-State batteries | Medium | Sprint-2 |
| Sustainable Mobility Research Analyst | Battery & Range Analysis | USN-7 | As a research analyst, I can compare battery technologies by sales and average battery cycle life, so that I can evaluate technology trends. | Dual-axis chart shows sales bars with a cycle life line for each battery type | High | Sprint-2 |
| Sustainable Mobility Research Analyst | Battery & Range Analysis | USN-8 | As a research analyst, I can view the driving range distribution by vehicle category, so that I can assess range improvements. | Box plot shows the range (km) spread for every vehicle category | Medium | Sprint-2 |
| Sustainable Mobility Research Analyst | Charging Performance | USN-9 | As a research analyst, I can compare maximum DC charging power with energy efficiency, so that I can study charging performance across categories. | Scatter plot shows charging kW against Wh per km with average reference lines | Medium | Sprint-3 |
| Sustainable Mobility Research Analyst | Environmental Impact | USN-10 | As a research analyst, I can see CO2 saved per year by market region and drivetrain, so that I can support sustainability and policy recommendations. | Regional chart shows CO2 saved (tons per year) for AWD, FWD and RWD in every region | High | Sprint-2 |
| Dashboard User (All Roles) | Interaction & Filtering | USN-11 | As a dashboard user, I can filter and highlight by vehicle category using the shared legend, so that I can focus on one segment across all charts. | Selecting a category updates every chart on the dashboard | Medium | Sprint-3 |
| Dashboard Administrator | Data Management | USN-12 | As a dashboard administrator, I can connect and refresh the EV dataset, so that dashboards always show current data. | Refreshing the data source updates all worksheets without errors | High | Sprint-1 |